Curtis V. McCully

Experience

2018-Present **Las Cumbres Observatory**, *Senior Astrodata Scientist.*
In charge of all pipelines to analyze data taken by the observatory, approximately 3 million images per year and hundreds of spectra per month. Continued my own scientific programs on peculiar supernovae and gravitational wave follow up. Project Manager for the Scalable Cyber Infrastructure for Multi-Messenger Astronomy (SCIMMA) collaboration. We develop collaboration and coordination tools for the astronomical community to ensure that we can rapidly identify and follow gravitational-wave electromagnetic counterparts. I manage the development of the project across multiple institutions. I am principal investigator (PI) for the Treasure Map, a tool designed to also coordinate the astronomical community to search for electromagetic counterparts of gravitational wave events. This grant is specifically to bring the Treasure Map to professional cyber-infrastructure standards so that it can be used for future gravitational wave observing runs.
>269 refereed journal publications: A full publication list is available at `https://cmccully.github.io/publicationlist.pdf`

2021–2023 **Mount Vernon Nazarene University**, *Adjunct Faculty.*
Taught Astronomy 1 and Intro to Data Science

2018–2020 **Westmont College**, *Adjunct Faculty.*
Taught Classical Mechanics, Waves and Optics, and Quantum Mechanics I & 2

2015–2018 **Las Cumbres Observatory**, *Network Operations Scientist.*
Data reduction and analysis, thousands of images per night across 20 telescopes. Integration of the pipeline with other observatory software systems.

2014–2018 **Las Cumbres Observatory/UCSB**, *Postdoctoral Scholar.*
Data reduction and analysis for the Supernova Key Project at LCO. We currently have observations of more than 650 supernovae, with more than 150,000 images. I mentored three graduate students and several undergraduate students. I led studies on supernovae, gravitational lensing, and the first discovered kilonova (see `https://www.cnet.com/news/gravitational-waves-kilonova-explosion-nasa-ligo/`).

2008–2014 **Graduate Student**, *Rutgers University*, Department of Physics and Astronomy.
Studied peculiar supernovae: Spectroscopic and image data reduction in Python and C for a variety of telescopes including the Hubble Space Telescope. I also designed a new efficient computational framework that makes it possible to include hundreds of components in gravitational lens models.

Technical Skills

Open Source Projects
- Image Data Reduction Pipeline for LCO data: `github.com/lcogt/banzai`
- Fast Cosmic Ray Rejection: `github.com/astropy/astroscrappy`
- Image subtraction pipeline for HST: `github.com/cmccully/snhst`
- AstroPy Contributor `github.com/astropy/astropy`

Languages Python, Cython, IDL, Matlab, Mathematica, C, C++, Java, VB.net, VBA, TCL

Technologies AWS, Kubernetes, Django, Celery, MySQL, SQLite, RabbitMQ, ElasticSearch, MS Access, ArcGIS, Docker, Rancher, IRAF/PyRAF

Analysis Experience Image analysis, Monte Carlo Markov Chains, Maximum Likelihood Analysis (Chi-squared minimization), Parallel Computing (OpenMP, Multiprocessing, Beowulf Cluster Computing), Robotic Telescope Operations, Data Visualization, Fourier analysis, Neural Networks, Principle Component Analysis, Gaussian Processes

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Awards, Fellowships, and Grants

2023 Principal Investigator on NSF AST-2308113 CDS&E: WoU-MMA: The Treasure Map

2012 Principal Investigator Cycle 20 HST-GO-12973

2011-2012 Excellence in Mentoring Fellow

2011 AAS Chambliss Astronomy Achievement Student Award (Graduate)

2009–2010 Richard J. Plano Outstanding Teaching Assistant Award

2008-2009 Graduate Assistance in Areas of National Need (GAANN) Fellow

2008 AAS Chambliss Astronomy Achievement Student Award (Undergraduate)

Teaching and Mentoring

2008–Present Taught Astronomy I and Intro to Data Science at MVNU. Taught Classical Mechanics, Waves and Optics, and Quantum Mechanics I and II at Westmont College. At Rutgers, I was the Teaching Assistant for two introductory physics for underprepared engineering students. Detailed reviews are available upon request. I have mentored two postdocs, seven graduate students from UCSB and more than a dozen undergraduate students from Rutgers, UCSB, and Westmont College.

Service and Outreach

2025– Science and Technology Advisory Committee for Gemini Telescopes

2023 NASA SOMA Reviewer

2016– Astronomy on Tap Coordinator/Presenter

2014– Referee for MNRAS and ApJ

2016–2019 LCO Seminar Coordinator

2015–2016 LCO Journal Club Coordinator

09,11,13,14 Rutgers Graduate Student Life Committee Student Representative

2012 Rutgers Telescope Allocation Committee for the Southern African Large Telescope

Education

2004–2008 **Bachelor of Science**, *Southern Nazarene University*, Bethany, OK.
Summa Cum Laude. Major in Physics, Minor in Computer Science

2008–2014 **Ph.D.**, *Department of Physics and Astronomy, Rutgers University*, Piscataway, NJ.

References

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  keeton@physics.rutgers.edu
- Prof. Saurabh Jha
  Rutgers University
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